Rajalakshmi Engineering College

FUNDAMENTALS OF DATA SCIENCE LAB

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EXPERIMENT 9

Understand Matplotlib library use cases in Data Science

Experiment 9: Understand Matplotlib library use cases in Data Science

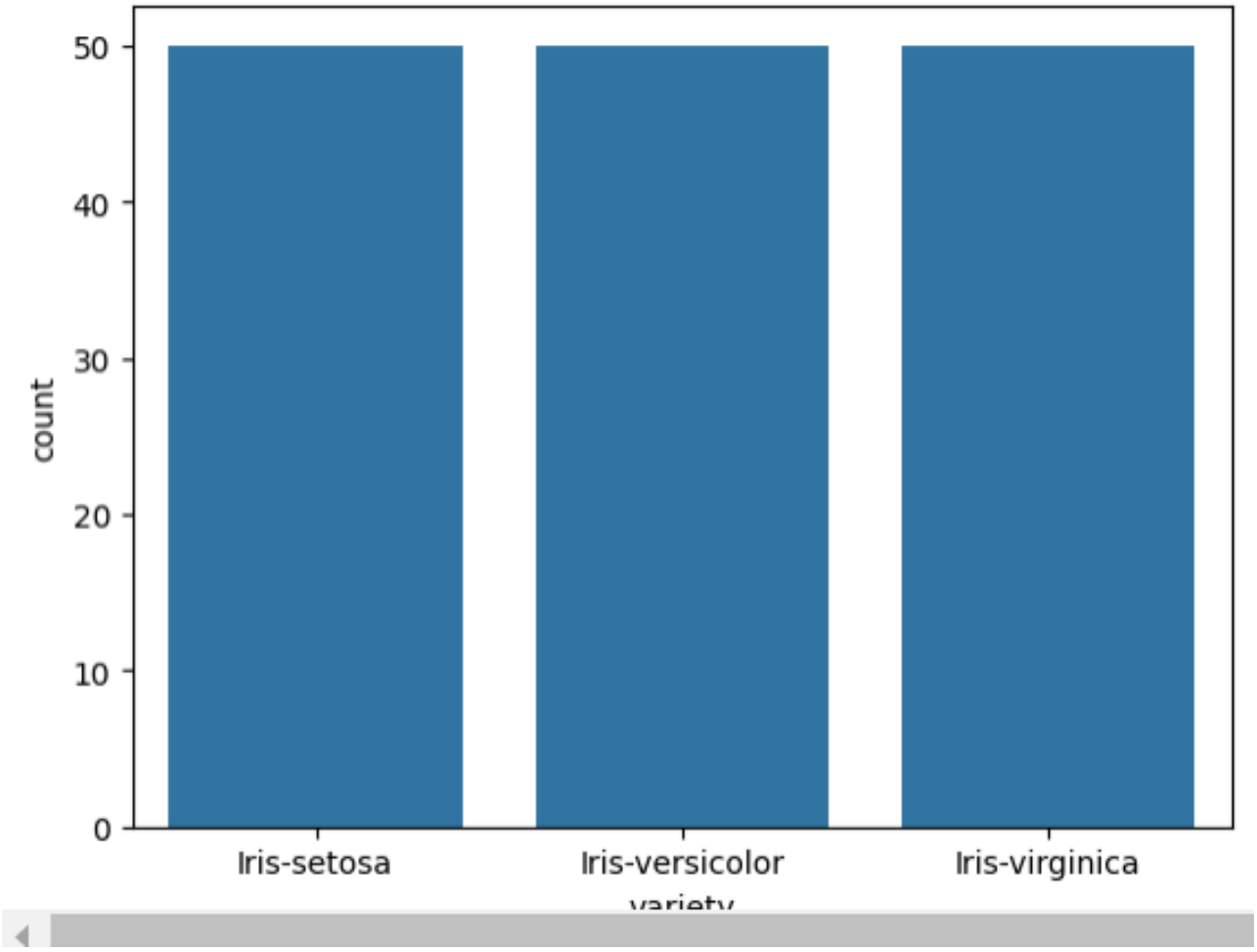
Aim

To demonstrate Matplotlib use cases in data science using the Iris dataset.

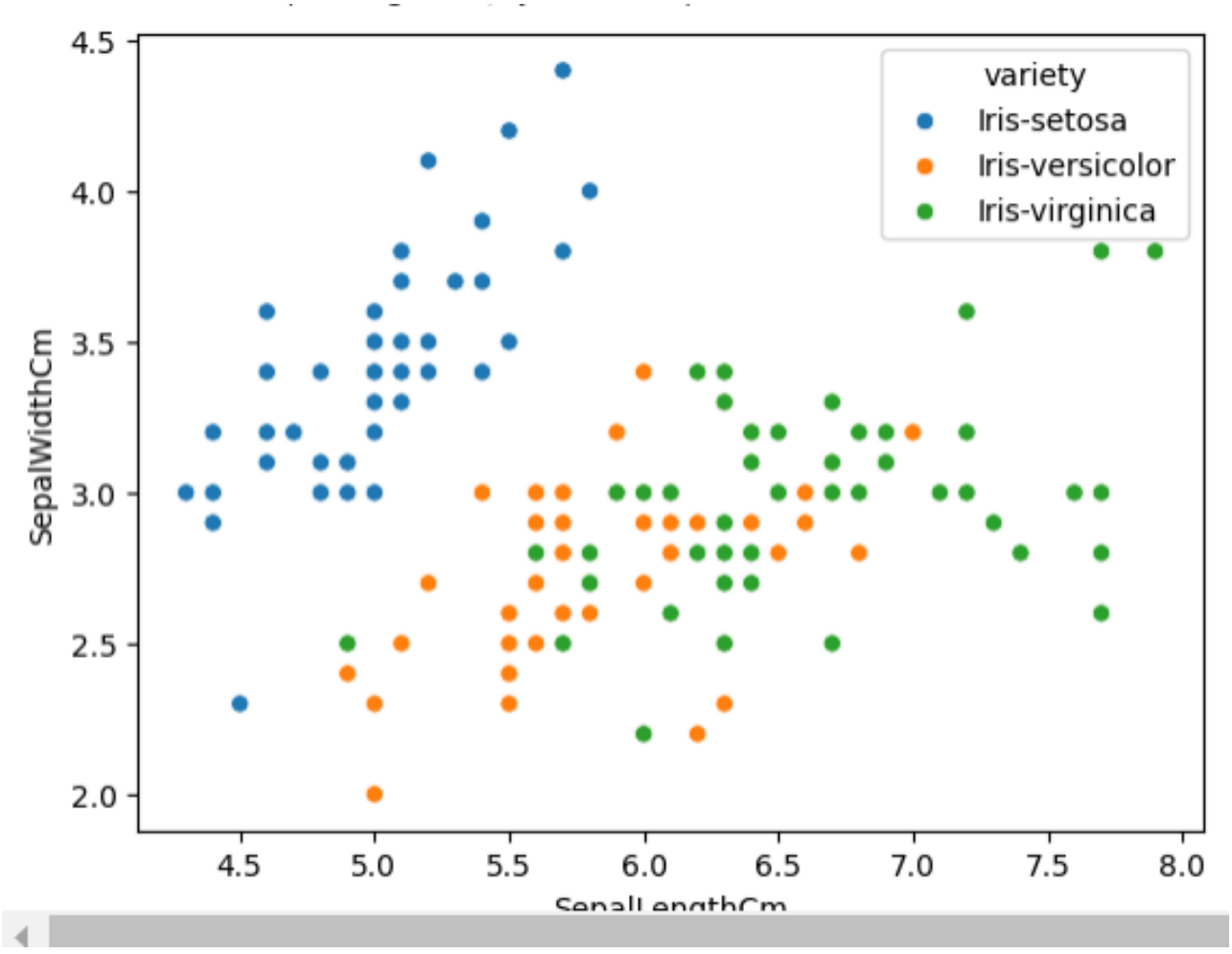
Source Code

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
data=pd.read_csv('/content/Iris_Dataset.csv')
data
... ... ... ... ... ... ...
data.info()
data.describe()
data.value_counts('variety')
sns.countplot(x='variety',data=data,)
plt.show()
dummies=pd.get_dummies(data.variety)
FinalDataset=pd.concat([pd.get_dummies(data.variety),data.iloc[:,[0,1,2,3]]],axis=1)
FinalDataset.head()
sns.scatterplot(x='SepalLengthCm',y='SepalWidthCm',hue='variety',data=data,)
sns.scatterplot(x='PetalLengthCm',y='PetalWidthCm',hue='variety',data=data,)
sns.pairplot(data,hue='variety',height=3);
plt.show()
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalLengthCm').add_legend();
plt.show();
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalWidthCm').add_legend();
plt.show();
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalLengthCm').add_legend();
plt.show();
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalWidthCm').add_legend();
plt.show();
```

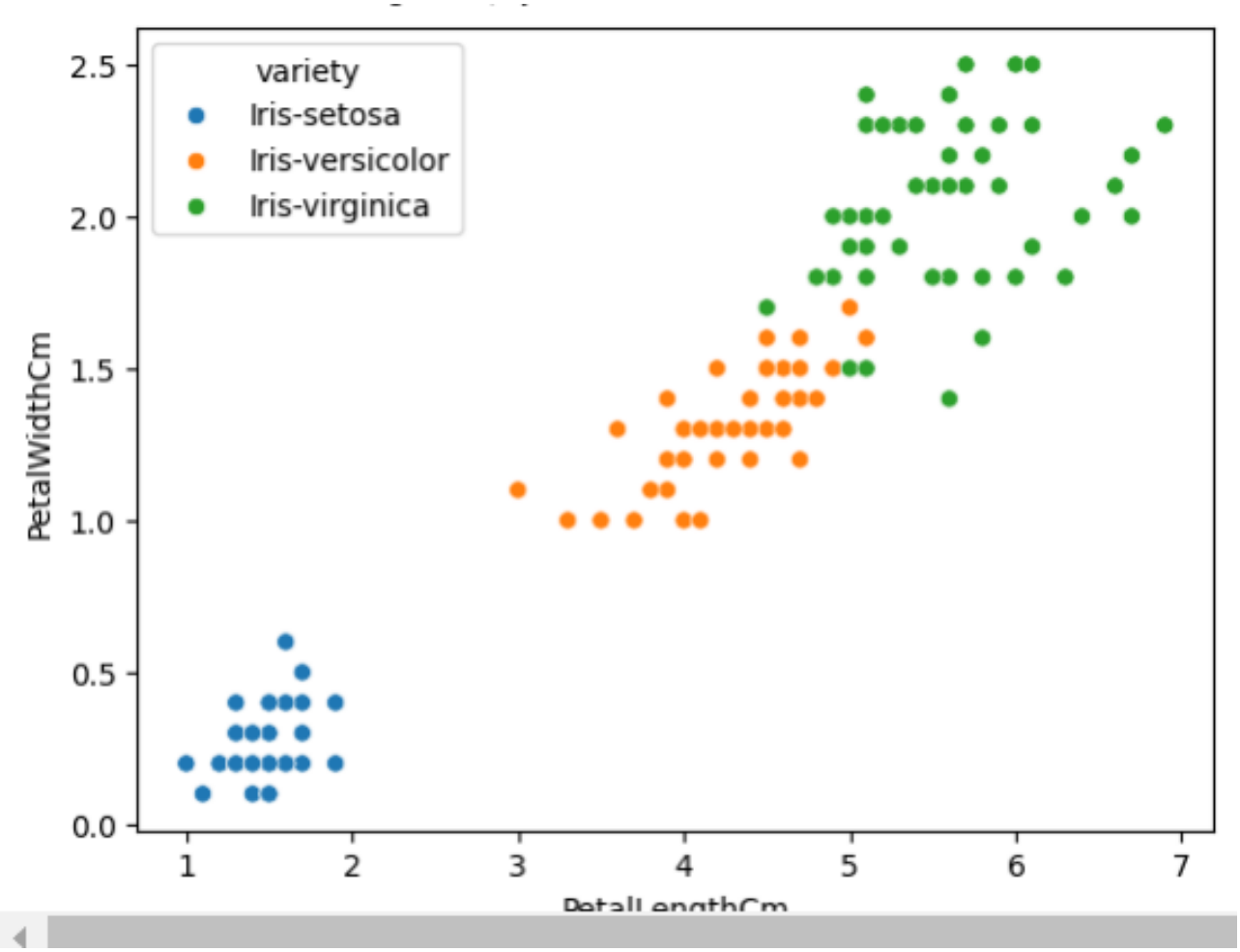
Output - Experiment 9



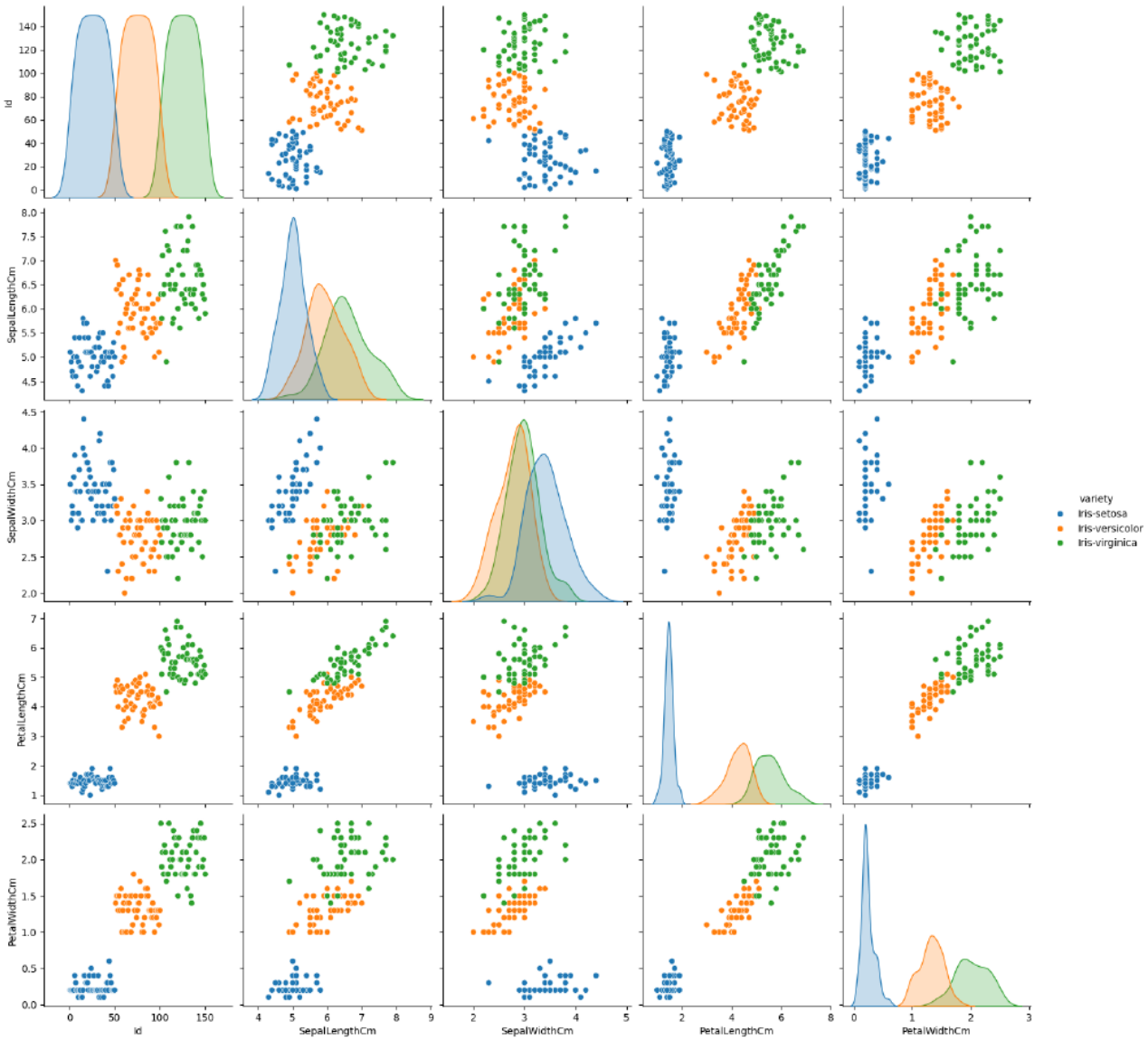
Output - Experiment 9



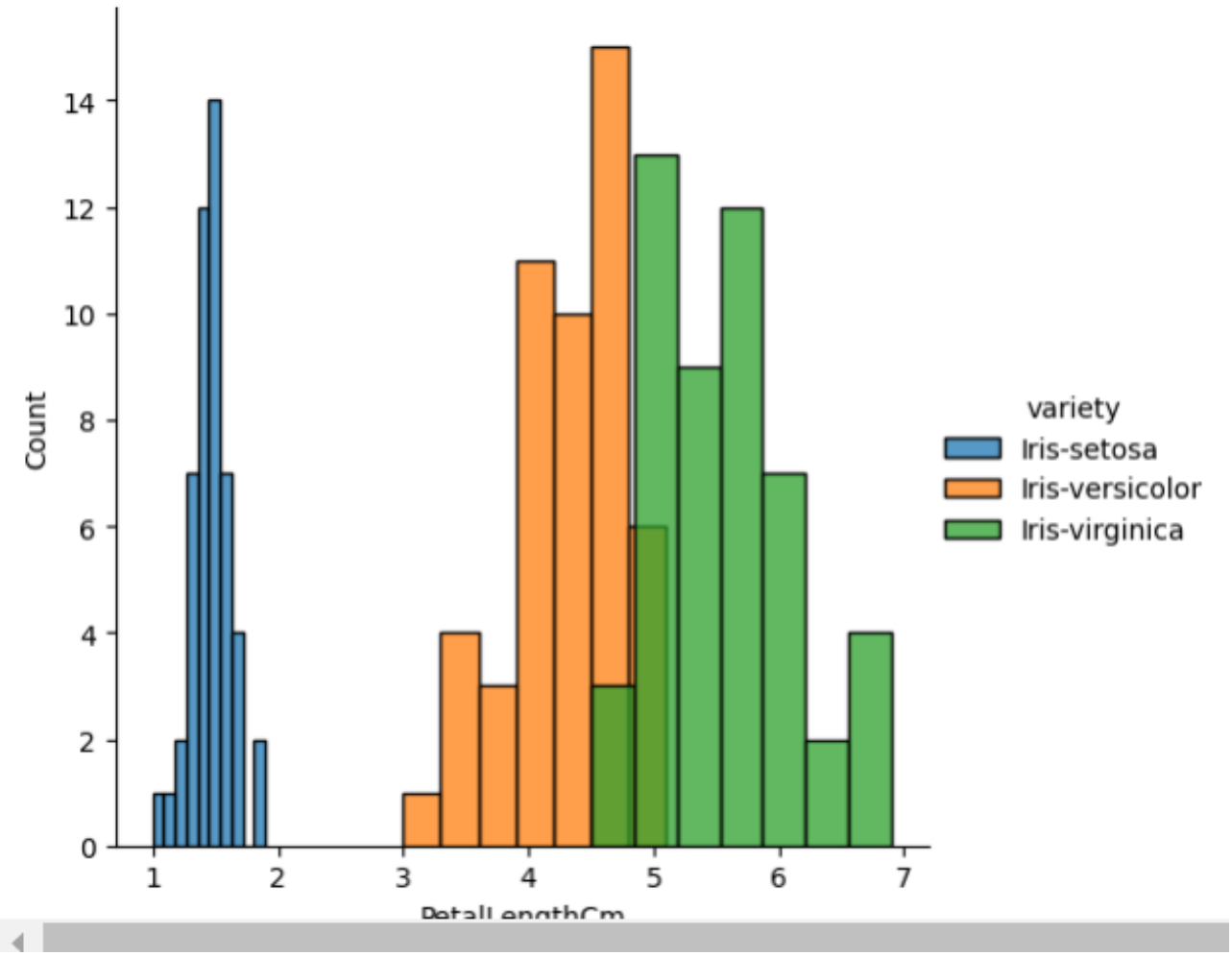
Output - Experiment 9



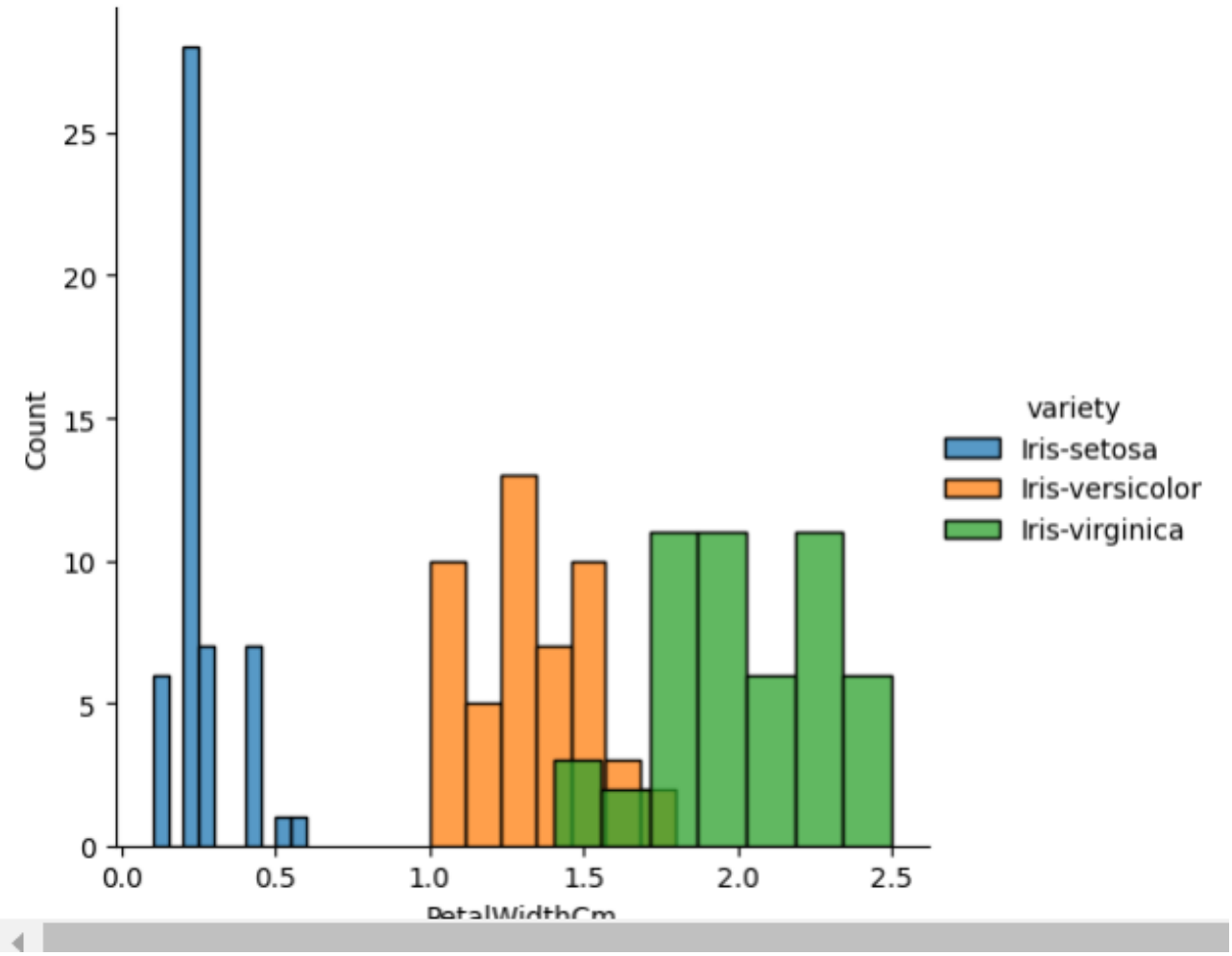
Output - Experiment 9



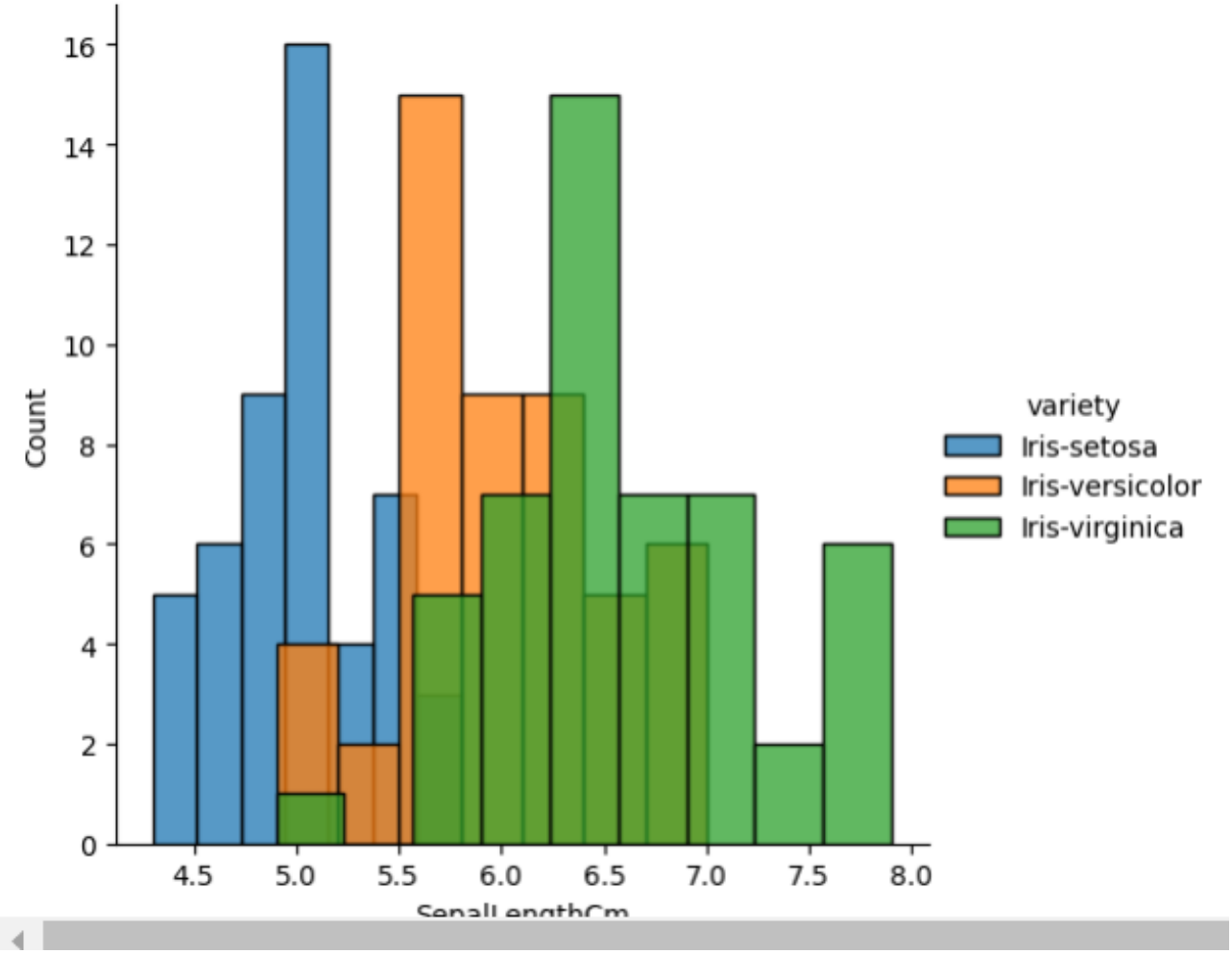
Output - Experiment 9



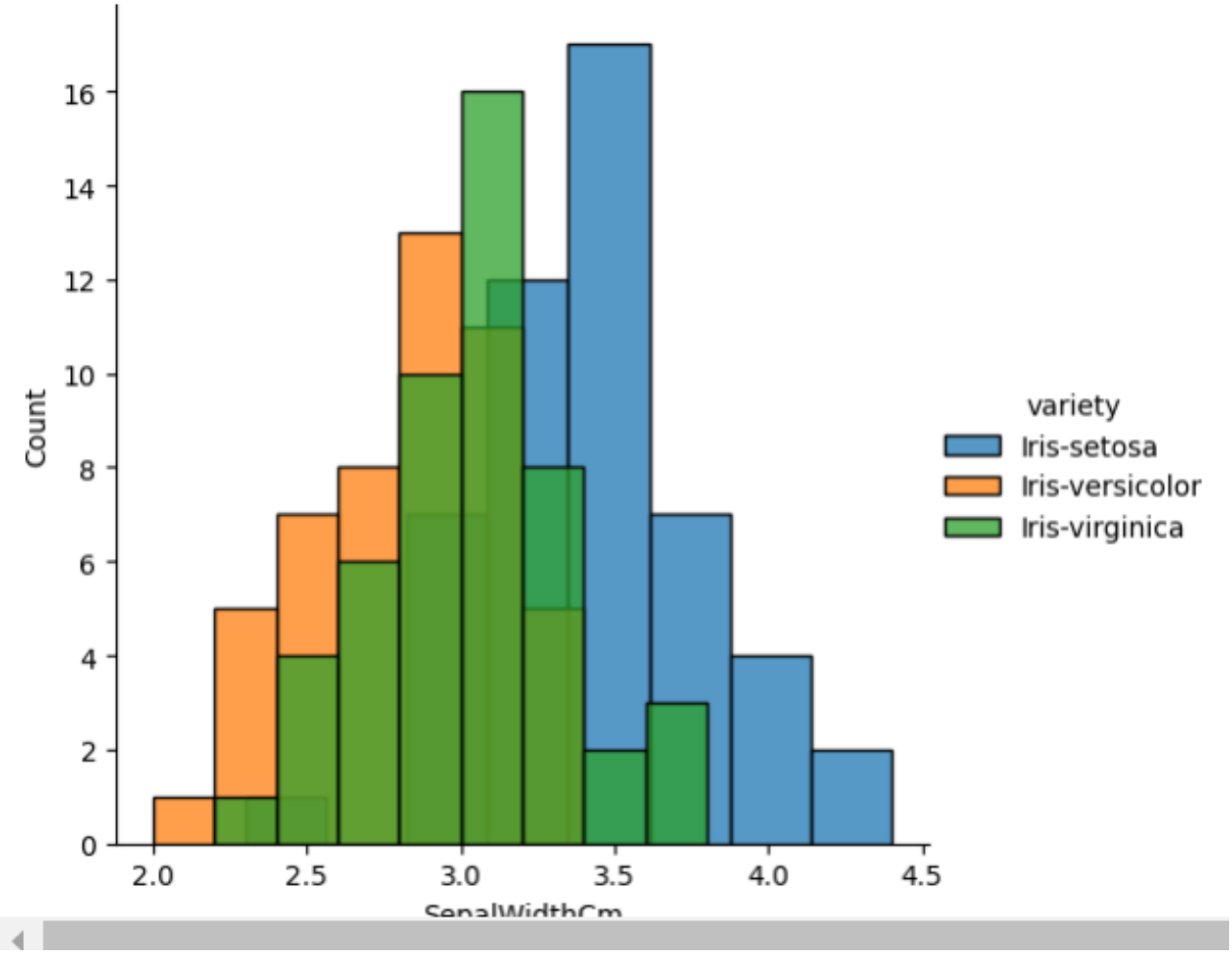
Output - Experiment 9



Output - Experiment 9



Output - Experiment 9



Output - Experiment 9

Id SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm variety
0 1 5.1 3.5 1.4 0.2 Iris-setosa
1 2 4.9 3.0 1.4 0.2 Iris-setosa
2 3 4.7 3.2 1.3 0.2 Iris-setosa
3 4 4.6 3.1 1.5 0.2 Iris-setosa
4 5 5.0 3.6 1.4 0.2 Iris-setosa
145 146 6.7 3.0 5.2 2.3 Iris-virginica
146 147 6.3 2.5 5.0 1.9 Iris-virginica
147 148 6.5 3.0 5.2 2.0 Iris-virginica
148 149 6.2 3.4 5.4 2.3 Iris-virginica
149 150 5.9 3.0 5.1 1.8 Iris-virginica
150rows×6columns
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):

Result - Experiment 9

Matplotlib library use cases were demonstrated through data visualizations.